EE2001 Electrical Circuits I:

Continuation to:

Circuit Variables

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Outline



Continuation to: 3. The International System of Units (SI)

4. Circuit Analysis

5. Voltage and Current

6. The Ideal Basic Circuit Element

7. Passive Sign Convention

8. Power and Energy

The International System of Units (SI)

- Engineers **compare** theoretical results to experimental results and compare competing engineering designs **using quantitative measures**.
- Modern engineering is a **multidisciplinary profession,** in which teams of engineers work together on projects and they can **communicate their results** in a meaningful way if they all use **the same units of measure**.
- **The international system of units (SI)** is a unifying system of units used by many societies and engineers throughout the world.
- The SI units are based on **seven defined base quantities**. One of them is **electric current**.
- The other quantities are **derived** from the **base quantities**.
- In case of **too small or too large quantities**, **standard prefixes** corresponding to **powers of 10** are applied to the basic unit. Engineers often use only the **prefixes for powers divisible by 3** (mili (10^{-3}), micro (10^{-6}), nano (10^{-9}), pico (10^{-12}) and
- kilo (10^{3}), mega (10^{6}), giga (10^{9}), tera (10^{12}) are mostly used)

The International System of Units (SI)

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TABLE 1.1 The International System of Units (SI)

Quantity	Basic Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	degree kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

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The International System of Units (SI)

TABLE 1.2 Derived Units in SI

Quantity	Unit Name (Symbol)	Formula
Frequency	hertz (Hz)	s^{-1}
Force	newton (N)	$kg \cdot m/s^2$
Energy or work	joule (J)	$N \cdot m$
Power	watt (W)	J/s
Electric charge	coulomb (C)	$A \cdot s$
Electric potential	volt (V)	J/C
Electric resistance	ohm (Ω)	V/A
Electric conductance	siemens (S)	A/V
Electric capacitance	farad (F)	C/V
Magnetic flux	weber (Wb)	$V \cdot s$
Inductance	henry (H)	Wb/A

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The International System of Units (SI)

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TABLE 1.3 Standardized Prefixes to Signify Powers of 10

Prefix	Symbol	Power
atto	a	10^{-18}
femto	f	10^{-15}
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
deka	da	10
hecto	h	10^{2}
kilo	k	10^{3}
mega	M	10^{6}
giga	G	10^{9}
tera	T	10^{12}

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3 The International System of Units (SI)

- Some commonly used SI examples from our field:
  - **kilovolt (kV)**, **milivolt (mV)** of voltage
  - **kilowatt (kW)**, **megawatt (MW)**, **gigawatt (GW)** of electrical power
  - **kilowatt-hour (kWh)**, **megawatt-hour (MWh)**, **gigawatt-hour (GWh), terawatt-hour (TWh)** of electrical energy
  - **kiloohm (kΩ)** and **miliohm (mΩ)** for resistance value
  - **milihenry (mH)** for inductance value
  - **mikrofarad (µF)**, **nanofarad (nF)** and **picofarad (pF)** for capacitance value

Circuit Analysis

Where circuit analysis fits within the whole topics of circuit design?

- All engineering designs **begin with a need**, (e.g. **improve** on an **existing** design or something **brand-new**).
- A careful assessment of the need results in **design specifications**, which are measurable characterisics of a proposed design. Design specifications allow us to assess whether or not the design acutally meets the need.
- A **concept** for the design comes next based on education and experience. The concept may be realized as a **sketch**, as a **written description** or in some other form.
- Often the next step is to **translate** the concept into a **mathematical model**.
- Commonly used **mathematical model** for **electrial systems** is a **circuit model**.

4 Circuit Analysis

- The **circuit models** which describe electric circuits will be composed of **circuit components**.
- These circuit components will be **ideal circuit components** which constitute **mathematical models for the actual circuit elements**.
- It is important for the **ideal** circuit component used in a circuit model to represent the behavior of the **actual** electrical component to an **acceptable degree of accuracy**.
- The tools of **circuit analysis** are **mathematical techniques** which are used **to predict the behavior of** the circuit and its components.
- A **comparison** between the **desired behavior**, from the **design specifications**, and the **predicted behavior**, from **circuit analysis**, may lead to **refinements** in the circuit model and its **ideal circuit elements**.
- Once the **desired** and **predicted** behavior are **in agreement**, a **physical prototype** can be constructed.

5 Voltage and Current

- Electric charge is **bipolar**. It can be **positive** or **negative**.
- It exists in **discrete quantities**, which are multiples of the **charge of an electron**:

$$e = 1.6022 \times 10^{-19}\, C$$

- **Separation of charge** results **in electric potential**, which is referred to as **voltage** in this course.
- **Motion of charge** results in electric current.
- The SI unit for voltage is **Volt (V)**.
- The SI unit for current is **Ampere (A)**.

Voltage and Current

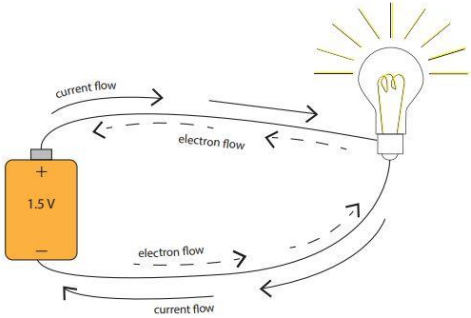
- Voltage (electric potential) is **electric potential energy per unit charge**:

$$v = \frac{dw}{dq}$$

- Electric current is **the rate of charge flow**:

$$i = \frac{dq}{dt}$$

v : the voltage in volts
w : the energy in joules
q : the charge in coulombs
i : the current in amperes
t : the time in seconds



- The voltage has a **polarity** and the current has a **direction**.
- Although in reality electrons are moving from negative terminal to positive terminal, the current flow direction is conventionally assumed from positive terminal to negative terminal.
- Although the charge is a discrete quantity, there are enormous number of charge and the current is treated as a continuous quantity.
- If two physically different circuit elements have the same current-voltage characteristics, their circuit models are identical and they are identical from the point of circuit analysis.

End of the Lesson (03.10.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist

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